AI-Driven Interactive Wine Cellar Management System with Transparent Display Integration

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***Abstract*—The premium home appliance sector is increasingly prioritizing artificial intelligence (AI) and user experience (UX) over mere hardware specifications. In the context of luxury wine preservation, existing commercial cellars remain passive climate-control units, while third-party software solutions suffer from high interaction friction and a lack of physical-digital synchronization. To address these bottlenecks, this paper presents an AI-driven, transparent-display-integrated wine cellar management system designed for high-end collections. Our system introduces a robust Vision AI pipeline for automated label recognition and spatial mapping, effectively eliminating manual inventory tracking even for flexibly stacked bottles. Furthermore, we implement a privacy-preserving, on-device Multi-Agent Small Language Model (SLM) architecture. By leveraging a quantized 4-bit base model with dynamic Low-Rank Adaptation (LoRA) swapping on a low-power Neural Processing Unit (NPU), the system generates persona-driven expert dialogues, dynamic food pairings, and personalized daily briefings without violating the stringent thermal and memory constraints of the appliance. Ultimately, this architecture transforms the traditional wine cellar from a passive storage unit into an actively managed, hyper-personalized asset management ecosystem.**

***Index Terms*—Smart Home Appliance, Edge AI, Vision AI, Small Language Model (SLM), Low-Rank Adaptation (LoRA), Wine Management System, Human-Computer Interaction (HCI)**

TABLE I. *Role Assignments*

Roles	Name	Task description and etc.
User	Chiang Jun Han	Describe typical user scenarios and workflows. Identify usability problems and practical needs. Try prototypes and provide feedback.
Customer	Naomi K. Fossaberg	Define the project's purpose and desired outcomes. Prioritize requirements and features. Evaluate whether the final solution meets the agreed requirements.
Software developer	Enkhbold Myagmarjav	Design and implement software components. Write and review code. Test functionality and fix bugs.
Development manager	Sungbin Bae	Plan tasks, milestones, and deadlines. Coordinate collaboration and monitor progress. Identify risks and help resolve blockers.

I. INTRODUCTION

A. Motivation

In recent years, the premium home appliance market has undergone a fundamental paradigm shift, transitioning from pure hardware performance competition toward elevating user experience (UX) through Artificial Intelligence (AI) and the Internet of Things (IoT). High-end luxury product lines, such as LG SIGNATURE, incur substantial hardware manufacturing costs, thereby demanding software capabilities that deliver commensurate value to justify their premium positioning. Concurrently, the global expansion of the fine wine market has driven rapid demand for specialized home wine cellars which is driven by AI management. However, existing commercial wine cellars remain passive appliances limited to fundamental condition control (temperature, humidity regulation, anti-vibration, and so on), failing to satisfy the sophisticated digital management needs of modern fine wine collectors.

B. Problem Statement and Target Audience

The primary target demographic for luxury wine cellars consists of high-net-worth individuals, predominantly in their 40s to 60s. Although these collectors frequently possess extensive and valuable wine collections, current management paradigms introduce critical usability bottlenecks:

1) **High Interaction Cost and Friction:** Existing mobile-app-based wine management solutions force users to repeatedly handle cold bottles while operating small smartphone screens. This physical separation between the appliance and the management interface induces high cognitive load and interaction friction.
2) **Absence of Physical-Digital Synchronization:** Current wine cellars lack real-time synchronization between physical storage and digital databases. Consequently, users are required to perform manual inputs for every bottle movement, leading to severe tracking discrepancies and user abandonment over time.

C. Related Software and Limitations

An analysis of existing wine management software (e.g., Vivino, CellarTracker) and proprietary cellar tracking apps reveals distinct functional limitations when applied to high-end collections:

- **Absence of Spatial Awareness:** Most commercial applications function merely as flat digital ledgers or list-based inventories. They lack a robust spatial mapping mechanism that correlates a specific bottle's data with its exact physical coordinates (e.g., specific racks and rows) within the cellar, inevitably forcing users to manually search through shelves.
- **Static Data without Environmental Context:** Existing software operates completely independently of the cellar's internal hardware sensors. Consequently, they cannot factor real-time environmental data—such as multi-zone temperature differences—into their aging algorithms or serving readiness calculations.
- **Reactive and Monotone User Experience:** Current tools are inherently reactive, requiring users to explicitly query the database to check a wine's optimal drinking window or food pairing suitability. Furthermore, they rely on static text reviews, lacking the proactive intelligence and dynamic, expert-level engagement necessary to autonomously curate an immersive experience.

D. Proposed Solution and Key Contributions

To overcome these limitations, this paper proposes an AI-driven wine cellar management system integrated with a transparent display door, designed specifically for luxury appliances such as the LG SIGNATURE line.

The key contributions of this work are summarized as follows:

- **Vision AI and Automated Spatial Mapping:** Integrates a robust vision pipeline to extract label and vintage data. Upon insertion, an automated placement mechanism registers the bottle's location. This extends beyond rigid 2D grids, possessing the capability to accurately map bottles even when they are flexibly stacked on top of one another without discrete rack partitions.
- **On-Device Multi-Agent SLM Architecture:** Introduces a privacy-preserving, edge-AI conversational interface where multiple Small Language Models (SLMs), each embodying the persona of a renowned wine critic, engage in dynamic debates regarding the selected wine. To circumvent the severe memory constraints of home appliances, the architecture leverages a single quantized base model with dynamic Low-Rank Adaptation (LoRA) swapping, ensuring a latency-free and hyper-personalized interactive experience without relying on external cloud servers.
- **AI-Driven UX Enhancement:** Transcending basic touch interactions, the integrated AI engine drastically elevates the overall user experience by providing a highly versatile suite of intelligent services. By analyzing multidimensional data, the system proactively offers a broad spectrum of personalized management features and domain-specific insights, transforming passive wine storage into an actively managed, adaptive ecosystem.
- **Multi-Device Distributed Scalability:** Establishes a synchronized system architecture that seamlessly unifies multiple physical cellar units under a single user profile for distributed inventory management.

II. REQUIREMENTS SPECIFICATION

A. Functional Requirements

1) User Management:

- The system shall allow a user to create an account with the following mandatory information: Username, Email, and Password.
- The system shall allow a registered user to log in and out.
- The system shall allow an authenticated user to edit their profile information.

2) Inventory Mapping:

- The system shall automatically detect wine bottles placed inside the wine cellar using computer vision.
- The system shall extract wine label information from detected bottles using AI.
- The system shall automatically register newly detected bottles in the inventory.
- The system shall automatically associate each detected bottle with a physical location in the wine cellar.
- The system shall automatically update a bottle's location when it is moved within the wine cellar.
- The system shall assign a unique identifier to each physical wine bottle to prevent duplicate inventory records.
- The system shall allow an authenticated user to review and correct AI-extracted wine information before confirming registration.
- The system shall automatically detect stacked wine bottles.
- The system shall automatically remove a bottle from the inventory when it is removed from the wine cellar.
- The system shall allow an authenticated user to manually add a wine bottle to the inventory.
- The system shall allow an authenticated user to manually modify bottle information.
- The system shall allow an authenticated user to manually remove a wine bottle from the inventory.

3) Inventory Search & Visualisation:

- The system shall allow a user to search for a bottle by wine name, winery, region, vintage or grape variety.
- The system shall display the physical location of a selected bottle within the wine cellar.
- The system shall display a visual map of bottles stored in the wine cellar.

4) Environmental Monitoring:

- The system shall collect environmental data from the wine cellar sensors.
- The system shall store historic environmental measurements from the wine cellar.
- The system shall associate environmental conditions with individual bottle locations.
- The system shall notify users when temperature or humidity exceeds user-defined thresholds for a specified duration.
- The system shall notify users when sensors become disconnected or environmental data is outdated.

5) AI Recommendations:

- The system shall allow a user to input data regarding a meal and receive wine pairing recommendations from the user's current inventory.
- The system shall allow a user to select a specific wine from the inventory and generate complementary food pairing suggestions, including culinary recipes or food delivery menu categories.
- The system shall recommend bottles that have reached their optimal drinking window.
- The system shall notify a user when wines are approaching their optimal drinking window.
- The system shall generate personalised recommendations based on historical consumption patterns.
- The system shall allow a user to input specific constraints, such as a target budget, flavor profile, or event theme, to request a curated wine list.
- The system shall automatically generate and display a customized wine list that satisfies the user-defined parameters, utilizing both the existing cellar inventory and external databases for potential new purchases.
- The system shall recommend suitable storage locations for newly added bottles.
- The system shall recommend wines by SLM query.

6) Personalized Daily Wine Briefing:

- The system shall aggregate daily global wine news, vintage reports, and market trends from external web sources and APIs.
- The system shall personalize the aggregated news content by cross-referencing it with the user's specific cellar inventory, highlighting articles relevant to their owned regions, grapes, or producers.
- The system shall automatically curate and display a "Daily Wine Briefing" on the cellar's front display, including inventory-specific insights (e.g., wines reaching their peak, humidity stability).
- The system shall send a summarized version of the daily briefing as a push notification to the user's synchronized mobile application at a user-defined time.

7) Real-Time Market Value Integration:

- The system shall integrate with external global wine market APIs (e.g., Wine-Searcher) to retrieve real-time pricing data for the specific vintages stored in the cellar.
- The system shall calculate and visualize the real-time aggregated market value of the user's entire wine collection on the front display.
- The system shall provide graphical visualizations of historical price trends and value appreciation for individual bottles to assist the user in asset management.

8) Multi-Cellar Management:

- The system shall allow a user to manage multiple wine cellar units and wine cellars under a single account.
- The system shall synchronise inventory information across connected wine cellars.
- The system shall allow a user to search across all connected wine cellars.

9) Multi-Agent SLM Dialog System:

- The system shall integrate multiple distinct On-Device Small Language Models (SLMs), each configured with a unique persona reflecting the evaluation tendencies and grading philosophies of renowned wine critics (e.g., Robert Parker, Luca Maroni).
- The system shall automatically trigger a simulated multi-agent dialogue among the embedded SLMs whenever a new wine is registered to the cellar inventory or actively selected by the user via the touch display.
- The system shall generate and visualize interactive, chat-style debates on the front display, showcasing diverse analytical perspectives, tasting notes, and aging predictions for the selected wine.
- The system shall allow the on-device SLMs to continuously fine-tune their conversational outputs and recommendations by locally learning from the user's cellar database, including inventory history, consumption patterns, and individual preferences.
- **Base Model Initialization:** The system shall load a single, heavily quantized (e.g., 4-bit) base Small Language Model (SLM) into the on-device volatile memory (RAM) upon system startup.
- **Dynamic LoRA Swapping:** The system shall dynamically load and unload specific Low-Rank Adaptation (LoRA) weight matrices corresponding to the active persona during the multi-agent dialogue generation.
- **Modular Persona Update:** The system shall allow the addition or modification of critic personas via OTA (Over-The-Air) updates by simply downloading new, lightweight LoRA files (under 50MB) without altering the base model.

10) Mobile Application Integration:

- The system shall provide a companion mobile application that synchronizes real-time inventory and cellar status with the user's account.
- The system shall allow a user to remotely browse their wine inventory, search for bottles, and view AI recommendations via the mobile application.
- The system shall push real-time notifications to the mobile application regarding critical environmental alerts

(e.g., temperature fluctuations) and optimal drinking windows.
- The system shall allow a user to remotely adjust the temperature and humidity settings of individual cellar zones through the mobile application.
- The system shall allow a user to manually add, edit, or remove wine inventory data via the mobile application when away from the physical cellar.

B. Non-Functional Requirements

1) Performance:

- The system shall return inventory search results within 2 seconds for 95% of requests.
- The system shall update the displayed inventory within 10 seconds after a bottle had been added or removed.
- The system shall execute the LoRA adapter swapping and generate the initial dialogue response within 1.5 seconds to ensure a seamless, real-time user experience.

2) Resource:

- The total active memory footprint (VRAM/RAM) of the base SLM and the concurrently loaded LoRA adapters shall strictly not exceed 4 GB.

3) Reliability:

- The system shall preserve the submitted data so that it remains after application restart.
- The system shall recover the latest successfully stored inventory after an unexpected shutdown.
- The system shall automatically back up inventory data and support recovery following unexpected data loss.

4) Security:

- The system shall require authentication before a user can access inventory management functions.
- The system shall store passwords in encrypted form.
- The system shall encrypt personal user data during transmission.
- The system shall prevent users from accessing or modifying another user's wine inventory.

5) Privacy:

- The system shall be compliant with GDPR.
- The system shall process all SLM-related computations and user preference learning entirely on-device (Edge AI), guaranteeing that no personal consumption data or inventory logs are transmitted to external cloud servers.

6) Maintainability:

- The system architecture shall support replacement or upgrade of AI models without requiring modifications to the user interface.
- The system architecture shall support integration of additional sensor types without requiring changes to existing inventory management functionality.

7) Usability:

- The system shall allow new users to use it without consulting external documentation.
- The system shall indicate the authentication of the current user at all times.
- The system shall preserve unsaved user input in the event of application restart, so the user will not have to reenter it.

8) Energy Efficiency:

- The SLM inference process shall be optimized to utilize the low-power Neural Processing Unit (NPU), ensuring that the AI chat feature does not generate excessive heat that could impact the internal temperature control of the wine cellar.
- The system shall exclusively utilize a low-power Neural Processing Unit (NPU) for on-device SLM inference, strictly prohibiting the use of high-wattage GPUs to prevent excess thermal output.
- The system's computational module shall be thermally isolated, ensuring its heat dissipation does not alter the internal climate zones of the wine cellar.
- The SLM inference process shall be optimized to operate within the stringent Thermal Design Power (TDP) limits of the embedded NPU, ensuring zero interference with the cellar's anti-vibration and precision cooling mechanisms.